

CONSTRAINTS

USE DBMS1_2025

-----1) NOT NULL CONSTRAINT-----

```
CREATE TABLE PERSON(  
ID INT NOT NULL,  
FIRSTNAME VARCHAR(255) NOT NULL,  
AGE INT );
```

```
INSERT INTO PERSON VALUES(1, 'GEETA', NULL);
```

```
SELECT * FROM PERSON
```

--ERROR

```
INSERT INTO PERSON VALUES(NULL, 'MEETA', 30);
```

```
INSERT INTO PERSON VALUES(2, NULL, 30);
```

--Create a NOT NULL constraint on the "Age" column
--when the "Persons" table is already created.

```
ALTER TABLE PERSON  
ALTER COLUMN AGE INT NOT NULL;
```

--WHY ERROR?

--BECAUSE EXISTING TABLE HAVE A RECORD WITH NULL AGE

```
DELETE FROM  
PERSON  
WHERE AGE IS NULL
```

--NOW, NOT NULL CONSTRAINT ON AGE COLUMN

```
ALTER TABLE PERSON  
ALTER COLUMN AGE INT NOT NULL;
```

--ERROR

```
INSERT INTO PERSON VALUES(1, 'SHYAM', NULL)
```

---VALID

```
INSERT INTO PERSON VALUES(3, 'MEERA', 25)
```

-----2) CHECK CONSTRAINT-----

```
CREATE TABLE PERSONS (  
ID INT ,  
LASTNAME VARCHAR(255),  
FIRSTNAME VARCHAR(255),  
AGE INT CHECK (AGE >= 18)  
);
```

--ERROR

```
INSERT INTO PERSONS VALUES (1, 'PATEL', 'MEET', 12)
```

```
--VALID
```

```
INSERT INTO PERSONS VALUES (1, 'PATEL', 'JEET', 25)
```

```
DROP TABLE PERSONS
```

```
CREATE TABLE PERSONS (  
    ID INT ,  
    LASTNAME VARCHAR(255),  
    FIRSTNAME VARCHAR(255),  
    AGE INT  
);
```

```
--Create a CHECK constraint on the "Age" column  
--when the "Persons" table is already created.
```

```
ALTER TABLE PERSONS  
ADD CHECK (AGE>=18);
```

```
--ERROR
```

```
INSERT INTO PERSONS VALUES (1, 'SHAH', 'NISHIT', 12);
```

```
-----3) DEFAULT CONSTRAINT-----
```

```
CREATE TABLE PERSON_INFO (  
    ID INT ,  
    LASTNAME VARCHAR(255) ,  
    FIRSTNAME VARCHAR(255),  
    AGE INT DEFAULT 18  
);
```

```
INSERT INTO PERSON_INFO (ID, LastName, FirstName)  
VALUES (1, 'PATEL', 'SEEMA')
```

```
SELECT * FROM PERSON_INFO
```

```
--Create a DEFAULT constraint on the "Age" column  
--when the "PERSON_INFO" table is already created.
```

```
DROP TABLE PERSON_INFO
```

```
CREATE TABLE PERSON_INFO (  
    ID INT ,  
    LASTNAME VARCHAR(255) ,  
    FIRSTNAME VARCHAR(255),  
    AGE INT  
);
```

```
ALTER TABLE PERSON_INFO  
ADD CONSTRAINT df_Age  
DEFAULT 18 FOR AGE;
```

```
INSERT INTO PERSON_INFO (ID, LastName, FirstName)
VALUES (1, 'PATEL', 'SEEMA')
```

```
SELECT * FROM PERSON_INFO
```

```
-----4) UNIQUE KEY-----
```

```
CREATE TABLE PERSON_MASTER (
    ID INT UNIQUE,
    LASTNAME VARCHAR(255),
    FIRSTNAME VARCHAR(255) ,
    AGE INT
);
```

```
INSERT INTO PERSON_MASTER
VALUES(1, 'PATEL', 'BHOOMI', 19)
```

```
--ERROR
```

```
INSERT INTO PERSON_MASTER
VALUES(1, 'PATEL', 'BHOOMI', 19)
```

```
----
```

```
INSERT INTO PERSON_MASTER
VALUES(NULL, 'SHAH', 'KRISHNA', 19)
```

```
--ERROR
```

```
INSERT INTO PERSON_MASTER
VALUES(NULL, 'PATEL', 'KRISHNA', 19)
```

```
DROP TABLE PERSON_MASTER
```

```
CREATE TABLE PERSON_MASTER (
    ID INT,
    LASTNAME VARCHAR(255),
    FIRSTNAME VARCHAR(255) ,
    AGE INT
);
```

```
--Create a UNIQUE KEY on the "ID" column
--when the "PERSON_MASTER" table is already created
```

```
ALTER TABLE PERSON_MASTER
ADD UNIQUE (ID);
```

```
INSERT INTO PERSON_MASTER VALUES(2, 'PATEL', 'MITA', 19)
```

```
--ERROR
```

```
INSERT INTO PERSON_MASTER VALUES(2, 'PATEL', 'MITA', 19)
```

```
DROP TABLE PERSON_MASTER
CREATE TABLE PERSON_MASTER (
    ID INT,
    LASTNAME VARCHAR(255),
    FIRSTNAME VARCHAR(255) ,
```

```
    AGE INT
);

--ADD UNIQUE CONSTRAINT ON MULTIPLE COLUMN
ALTER TABLE PERSON_MASTER
ADD UNIQUE (ID, LASTNAME);

--OR
ALTER TABLE Persons
ADD CONSTRAINT UC_Persons UNIQUE (ID, FirstName);

-----5) PRIMARY KEY CONSTRAINT-----

CREATE TABLE CUSTOMER (
    ID INT PRIMARY KEY,
    LastName varchar(255),
    FirstName varchar(255),
);

INSERT INTO CUSTOMER VALUES(1, 'PATEL', 'BHOOMI')

--error (as ID is not unique)
INSERT INTO CUSTOMER VALUES(1, 'PATEL', 'SITAL')

--ERROR (as ID is NULL)
INSERT INTO CUSTOMER VALUES(NULL, 'PATEL', 'SITAL')

---Create a PRIMARY KEY on the "ID" column
--when the "CUSTOMER" table is already created.

DROP TABLE CUSTOMER

CREATE TABLE CUSTOMER (
    ID INT not null,
    LastName varchar(255),
    FirstName varchar(255),
);

--ID COLUMN MUST BE NOT NULL
ALTER TABLE CUSTOMER
ADD PRIMARY KEY (ID);

-----6) FOREIGN KEY CONSTRAINTS-----

CREATE TABLE CUSTOMER_DATA (
    ID INT PRIMARY KEY,
    LastName varchar(255),
    FirstName varchar(255),
);

CREATE TABLE ORDERS (
```

```
ORDERNO INT NOT NULL,  
CUST_ID INT FOREIGN KEY REFERENCES CUSTOMER_DATA(ID) );
```

--ERROR

```
INSERT INTO ORDERS VALUES(1,5);
```

--FIRST INSERT RECORD IN CUSTOMER_DATA TABLE(parent table)

```
INSERT INTO CUSTOMER_DATA VALUES(5, 'PATEL', 'SEEMA')
```

--NOW YOU CAN INSRT RECORD IN ORDER TABLE WITH CUSTOMER ID 5

```
INSERT INTO ORDERS VALUES(1,5);
```

----Create a FOREIGN KEY on the "CUST_ID" column

--when the "Order" table is already created.

```
DROP TABLE ORDERS
```

```
CREATE TABLE ORDERS (  
ORDERNO INT NOT NULL,  
CUST_ID INT );
```

```
ALTER TABLE ORDERS
```

```
ADD FOREIGN KEY (CUST_ID) REFERENCES CUSTOMER_DATA(ID);
```